

IMMUNOLOGIC ASSAYS IN THE POST-COVID ERA

삼성서울병원 진단검사의학과 양정석

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Landscape of Respiratory Pathogens

• The Big Three

SARS-CoV-2 (COVID-19)

- Now endemic with year-round circulation and variant-driven peaks
- Omicron sublineages (XFG, NB.1.8.1, BA.3.2) continue immune escape
- Severe disease risk persists for elderly, immunocompromised & comorbid patients
- Population immunity has decoupled infection rate from severe mortality

Influenza A & B

- Returned to traditional seasonal patterns after COVID disruptions
- H3N2 and H1N1pdm09 drive severe seasonal disease
- Influenza B/Yamagata lineage is now extinct; vaccines shifted to trivalent
- New A(H3N2) clades (e.g., Subclade K) continue to challenge vaccine efficacy

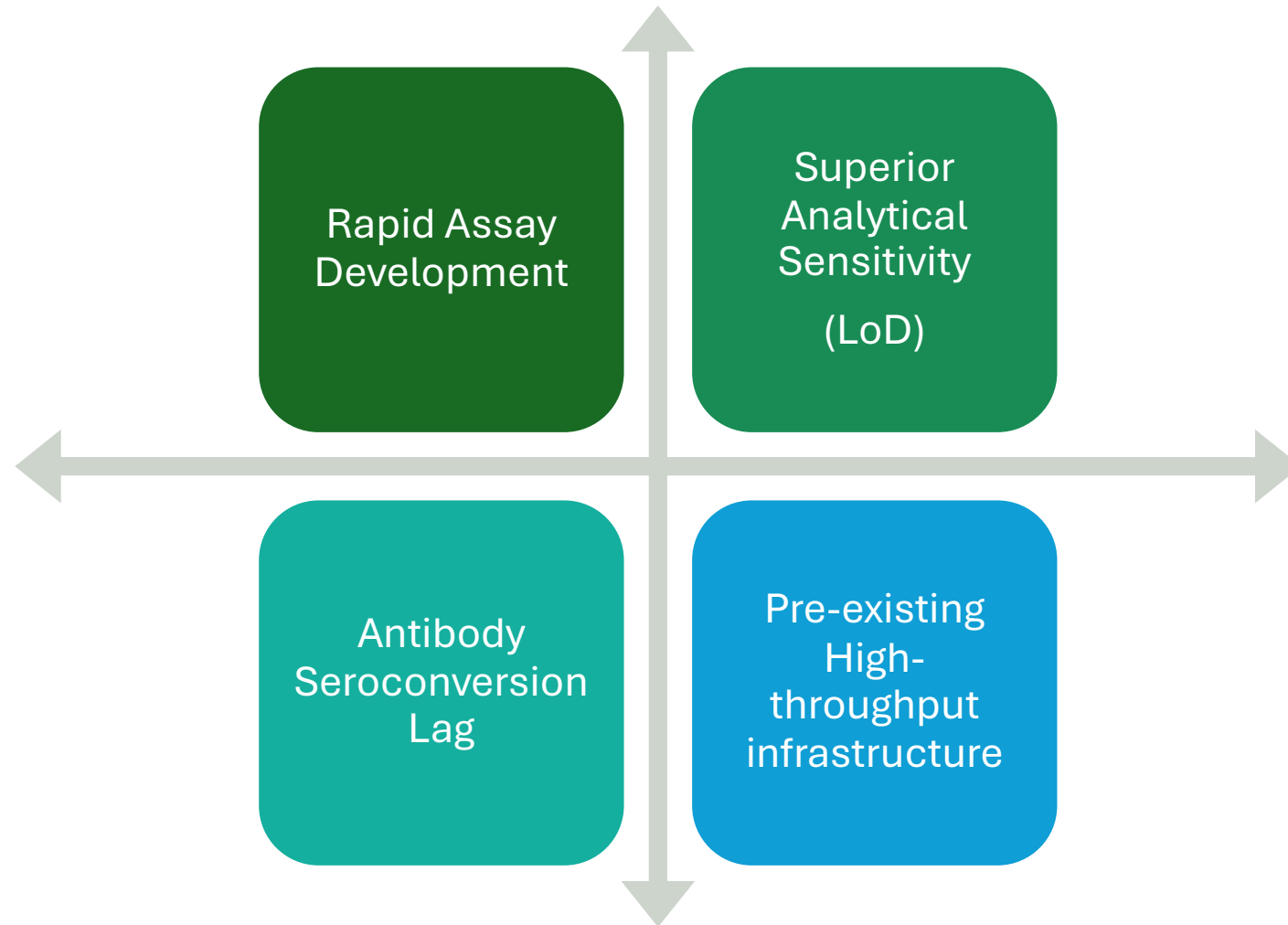
Respiratory Syncytial Virus (RSV)

- Now recognized as a serious threat beyond pediatrics — adults >65 at high risk
- New RSV vaccines for older adults alter epidemiological footprint
- Maternal vaccines & nirsevimab protect infants from severe disease
- Highly seasonal; co-circulation with Flu & COVID creates diagnostic challenges

Endemic and Emerging Pathogens

- **Rhinovirus / Enterovirus** — most common cause of the common cold; year-round circulation
- **Human metapneumovirus (hMPV)** — clinically mirrors RSV; key driver of bronchiolitis in late winter/spring
- **Parainfluenza (PIV 1–4)** — croup in children; severe LRTI in immunocompromised adults
- **Mycoplasma pneumoniae** — post-pandemic out-of-season surges; 'walking pneumonia'

Dominance of Molecular Methods



1. Rapid Assay Development

- PCR design within days of genome publication (data sharing)
 - immunoassay development requirements ; epitope mapping, antibody production (scaling)
- Requires only oligonucleotide primers/probes
 - c.f. Antibody development

2. Analytical Sensitivity

- PCR LoD: **10^2 - 10^3** viral copies/mL (exponential amplification)
- Ag LFIA LoD: **10^5 - 10^6** copies/mL, (no amplification)
- PCR detects presymptomatic shedders
 - c.f. Ag-Ab binding may not occur

3. Seroconversion Lag

- IgM / IgG detectable only 7-14 days post-symptom onset
- Serology becomes positive **AFTER** peak infectious period
- More applicable for retrospective surveillance & convalescent plasma selection

4. High-throughput infrastructure

- Core labs – Roche Cobas 6800/8800, Abbott m2000 ... etc.
 - open-channel systems
- EUA authorization allows immediate scale-up; ~x,xxx tests /day
- Automated extraction / amplification systems

Why Immunoassay Still Matters

- Ag tests require intact viral proteins

- Is the patient still contagious?

Determines Active Infectiousness

Point-of-care Utility

- LFIA; 10~15 min
- without Lab infrastructure
- immediate triage

- Low-cost Ag tests (first-line)
- Population-scale

Cost-effectiveness



Why Immunoassay Still Matters

- **Assessing Immune Status (& Vaccine Efficacy)**
 - only method to quantify humoral immune response after vaccination
 - Confirmation of Seroconversion (should a patient be given booster?)
- **Differentiation of Vaccine-induced vs Infection**
 - **Spike-targeted vaccines (N-protein, COVID-19)**
- **Post-infectious complications**
- **Seroepidemiology & True Prevalence Estimation**

Why Immunoassay Still Matters

- **Assessing Immune Status (& Vaccine Efficacy)**
- **Differentiation of Vaccine-induced vs Infection**
- **Post-infectious complications**
 - Post-influenza Guillain-Barre | Post-COVID MIS-C ; PCR negative
- **Seroepidemiology & True Prevalence Estimation**
 - asymptomatic /mild cases
 - seroprevalence studies required for accurate data on population exposure

Antigen vs Antibody

DIMENSION	ANTIGEN (Ag)	ANTIBODY (Ab)
Target	Viral/bacterial protein (direct)	Host-produced IgM/IgG/IgA
Detects	Active/acute infection	Immune response (past exposure or vaccine)
Detection Window	Symptomatic period; early infection	7–14 days post-onset; persists months/years
Clinical Use	Acute diagnosis, isolation decisions	Immune status, seroprevalence, post-infection dx
PCR Comparison	Complements PCR for infectiousness	PCR cannot assess immunity — only serology can
Limitation	Misses very low viral loads (presymptomatic)	Cannot diagnose acute infection

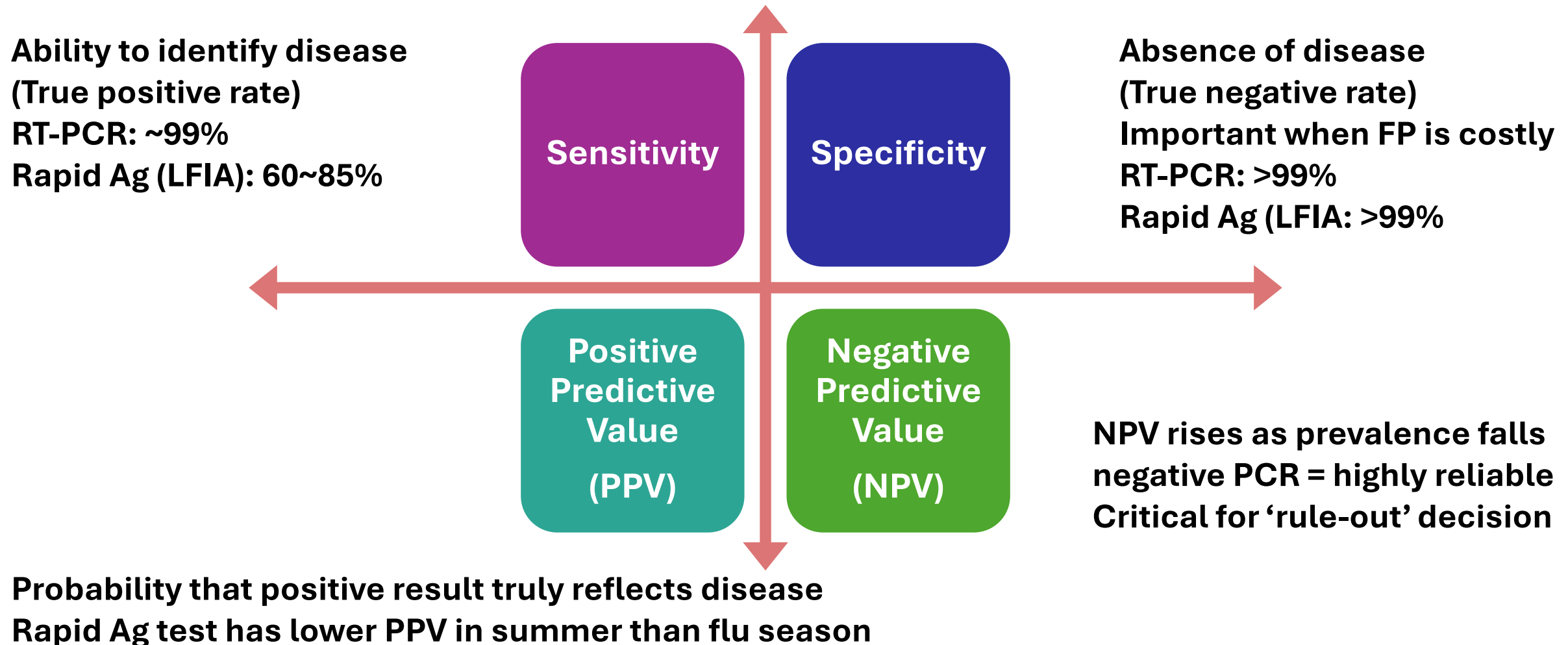
Types of Immunoassay Platforms

Platform	Sensitivity	TAT	Settings	Characteristics
Lateral Flow Immunoassay (LFIA)	Low-moderate	10-15 min	Point-of-care	No instrumentation, visual read out Rapid Ag detection High false-negative
Enzyme-linked Immunoassay (ELISA/EIA)	High	2-4 hrs	Laboratory	Gold standard for semiquantitative antibody measurement
Chemiluminescence Immunoassay (CLIA)	Very High	30-60 min	Core lab	Fully automated, high-throughput platforms Quantitative Ab titers High dynamic range (6-log)
Electrochemiluminescence (ECLIA)	Very High	18-30 min	Core lab	

Commercially Available Immunoassays

Platform	Product	Target	Clinical Use
LFIA	Abbott BinaxNOW COVID-19 Ag	SARS-CoV-2 Nucleocapsid (N) protein	Rapid (15-min) POC qualitative active infection detection
LFIA	Quidel QuickVue® Influenza A+B	Influenza A & B viral nucleoproteins	POC differentiation of Flu A vs B; guides immediate antiviral prescription
ELISA	EUROIMMUN Anti-SARS-CoV-2 IgG	IgG vs SARS-CoV-2 Spike S1 domain	Semi-quantitative post-infection/vaccination immune response assessment
ELISA	IBL International Influenza A IgG	IgG vs Influenza A antigens	Retrospective infection confirmation; vaccine efficacy research
CLIA	Abbott Alinity i SARS-CoV-2 IgG II	IgG vs Spike RBD (neutralizing Ab)	Quantitative vaccine-induced immunity monitoring; waning immunity tracking
CLIA	DiaSorin LIAISON® SARS-CoV-2 Ag	SARS-CoV-2 Nucleocapsid protein	High-throughput automated Ag detection; higher sensitivity vs LFIA
ECLIA	Roche Elecsys® Anti-SARS-CoV-2 S	IgG vs Spike RBD (BAU/mL)	Precise quantitative serology for humoral response; waning immunity studies
ECLIA	Roche Elecsys® Anti-SARS-CoV-2 N	Antibodies vs Nucleocapsid (N)	Differentiates natural infection (Anti-N+) from vaccine-only immunity (Anti-N-)

Interpretation of Assay Performance



Novel & Emerging Pathogens

- **Immune Monitoring**

- Torque Teno Virus (TTV)

- **Transplant Risk**

- Cytomegalovirus

- **Candidate Pathogens for pandemic**

- SARS-CoV-2 (XFG, NB1.8.1), Novel Influenza A variants

Torque Teno Virus (TTV)

- Anellovirus; ubiquitous, non-pathogenic DNA virus
 - everyone has TTV, lifelong latency
 - viremia level **inversely** correlates with immune competence
 - High TTV DNAemia → over-immunosuppression → risk of opportunistic infection (CMV, PCP)
 - Low TTV DNAemia → under-immunosuppression → risk of allograft rejection
 - Proposed as a real-time biomarker

Cytomegalovirus (CMV)

- Leading opportunistic infection in transplantation
- monitored by qPCR (CMV DNAemia)
 - antiviral therapy – ganciclovir, valganciclovir
- CMV-specific T-cell immunoassays
 - ELISPOT
 - QuantiFERON-CMV; monitoring for immunosuppression

Next Pandemic

- **Zoonotic Reservoir (Bats, swine, avian ...)**
 - broad RNA virus diversity & high human-animal interface
- **Antigenic novelty** – no pre-existing immunity
- **Respiratory transmission route**
 - droplet/aerosol dispersion enables $R_0 \gg 1$ before symptomatic detection
- **Surveillance Gap and Diagnostic Lag**

Next Pandemic Candidate Pathogens

- **Influenza A (H5N1 / H5Nx)**
 - H3N2 may cause entirely novel pandemic strain
- **Betacoronavirus (Novel Lineage)**
 - Multiple ACE2-binding strains, SARS-CoV-3 scenario
- **Nipah Virus (NiV); Bat-to-human (Southeast Asia)**
- **Enterovirus D68 / Novel EV; Periodic neuroinvasive outbreaks**

The role of Immunoassay in Pandemic

- **Serosurveillance** and True burden estimation
 - PCR confirms index cases
 - Serology reveals asymptomatic + undetected cases; population seroprevalence study/survey (Retrospective, biobanked serums)
- **Rapid Antigen Test** (Development); 6~10 weeks
 - epitope mapping post-genome publication
 - POC Ag testing allows mass screening without lab infrastructure

The role of Immunoassay in Pandemic

- **Vaccine immunogenicity** & Protective effect
 - Quantitative Ab assays (CLIA/ECLIA) defines immune threshold for protective effect – vaccine development
 - Neutralizing Ab titers – guide booster vaccination, timing
- Long-term immunity **surveillance**
 - longitudinal cohort serology tracking – National vaccination policy

Takeaways

- While molecular (PCR) assays dominate acute diagnostics, assessment of infectiousness and immunity still requires immunoassays
- Antigen assays are required for assessment of transmission risk
- Antibody assays are the only tools for quantitation of humoral immunity (infection vs vaccination)
- Comparison of assay function requires integration of prevalence (PPV, NPV)